Biomedical Data Science – Problem Set #3

Assigned: Monday, 10/12/2026
Due: Monday, 10/26/2026
Total: 100 Points

This problem set covers methods to account for generalized linear models, mixed effects modeling, time-event modeling, and data visualization with biomedical data sets. For this problem set, you are allowed to use the following libraries: *numpy, pandas, matplotlib, seaborn, scipy, statsmodel, sklearn,* and *lifelines* unless otherwise specified. Please submit your solutions to the Canvas site through Gradescope.

Problem 1 (25 pts). Consider the 'hospital charges' dataset (*insurance.csv*) from the previous problem set. As a reminder, the data columns contain patient hospital charges, age, body mass index (bmi), and smoking status. These data were derived from *Machine Learning with R* by Brett Lantz (Packt Publishing). For this problem, we are going to build a generalized linear model to predict hospital charges based on predictor variables (age, bmi, and smoking status).

(a) (10 pts) Justify an appropriate probability distribution for the response variable.

(b) (10 pts) Compare an ordinary linear regression model fit with a generalized linear model (using an appropriate link function given your observations in part a). Show a figure containing (i) an overlayed histogram plot comparing observed hospital charge distributions to model-predicted charge distributions for each model, and (ii) a scatterplot comparing predicted charges to residuals for each model.

(c) (5 pts) Compare MAE and RMSE for each model. Based on this analysis, which model does a better job at predicting hospital charges? Why do you think this is the case?

Problem 2 (20 pts). Consider a longitudinal study that investigated average reaction time (in milliseconds) for participants undergoing sleep deprivation. Days 0-1 were adaptation and training (T1/T2), day 2 served as baseline (B), and sleep deprivation started after day 2. These data are from Belenky, et al. (2003) and available on Canvas (*sleepstudy.csv*).

(a) (10 pts) Create a figure with subplots showing each participant's change in average reaction time over days. Include a linear regression line across days for each participant.

(b) (10 pts) Compare ordinary least squares regression (OLS), random intercept, random slope, and random intercept/slope models. Do participants differ in baseline reaction time? Do participants differ in how rapidly reaction time deteriorates?

Problem 3 (25 pts). Consider the dataset (*heartfailure.csv*) on Canvas, which contains features on demographic information and cardiovascular biomarkers. The 'time' column corresponds to the follow-up period (in days) and 'death_event' indicates if a patient died during the follow-up period. The data is from a paper by Chicco and Jurman (2020) and also available here.

(a) (10 pts) Compare Kaplan-Meier survival curves for patients with ejection fractions below 35% against those with ejection fractions 35% or higher.

(b) (10 pts) Predict survival curves using Cox proportional hazards regression in which you vary ejection fraction (25%, 40%, and 55%) while holding age and creatinine constant at their median values. Show a (i) hazard ratio forest plot for these three mortality predictors (age, serum creatinine, and ejection fraction) and (ii) survival curves for the three ejection fractions.

(c) (5 pts) What are the hazard ratios for a (i) 10 year increase in age, (ii) 0.5 mg/dL increase in creatinine, and (iii) 10% increase in ejection fraction?

Problem 4 (30 pts). Two relatively common vertebral disk pathologies are disk herniation and spondylolisthesis (see Figure 1 below). A dataset on Canvas (*vertebralcolumn.csv*) contains biomechanical attributes derived from the shape and orientation of the pelvis and lumbar spine as well as three diagnoses (normal: 100 patients, disk hernia: 60 patients, and spondylolisthesis: 150 patients). Recreate the following data visualization figure shown below Figure 1 (i.e. the one

containing parts A-D) using only *numpy, pandas, matplotlib, seaborn, and sklearn* libraries. Assume any random_state as 42 (recall Douglas Adams!).

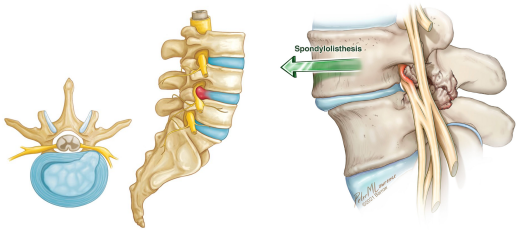


Figure 1. Examples of disk herniation (left) and spondylolisthesis (right)

